

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Distributed Energy Resources Templates

WECC DER Generation 13.5 MW 60 Hz,
WECC DER Generation 14.25 MW 50 Hz,
WECC DER Storage 13.5 MW 60 Hz,
WECC DER Storage 14.25 MW 50 Hz

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

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1 Introduction

This document describes the usage of the WECC Distributed Energy Resources (DER) model available in *PowerFactory*. The model implementation is based on the "DER_A" model description in [3].

The Electric Power Research Institute Inc. (EPRI) conducted an independent comparative analysis of the performance of the model as implemented in various commercial power system modelling software. The results of this analysis are presented in [2].

The model is used to represent distributed energy resources (DER) for generation or storage, for example within a network feeder, in an aggregated way, in which a single equivalent static generator represents the total output power and dynamics.

The distributed energy resources model "DER_A" defined by WECC is available in *PowerFactory* in form of the following templates:

- WECC DER Generation 13.5 MW 60 Hz
- WECC DER Generation 14.25 MW 50 Hz
- WECC DER Storage 13.5 MW 60 Hz
- WECC DER Storage 14.25 MW 50 Hz

In the templates for distributed generation, the parameter settings limit the power flow direction to power generation, while in the templates for distributed storage, power can flow in both direction (charging and discharging storage). Templates for use in 50 Hz power system and those for use in 60 Hz power systems differ in parameter settings of frequency-dependent behaviour (adapted to 50 Hz or 60 Hz systems respectively).

Chapters 2 and 3 describe how to use and configure the models in *PowerFactory*. In Chapter 4 the different dynamic models included in each template are described.

Note: The models are configured with a set of sample parameters. The user has the task to adjust static and dynamic model parameters (i.e. power rating, voltage level, reactive power limits and parameters of the dynamic models) to represent DER in specific power networks, being compliant with specific grid code requirements.

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2 Model Usage

Each of the templates consist of at least two network elements, the DER (*ElmGenstat*) and a terminal, and of a composite model which defines the dynamics of the DER model. The composite model contains the dynamic models built in DSL (*DIgSILENT* Simulation Language) and a voltage measurement device.

Note: The terminal serves as a measurement point for the voltage measurement device and should therefore not be deleted without changing the measurement point of the voltage measurement device accordingly.

The models are available as templates and are accessible to be used in a project via the icon *General Templates* in the drawing toolbox.

The models can be used for balanced RMS simulations. When using a fixed step size for the simulation, a typical value for the integration step size is a quarter up to a half of the period of the frequency under consideration. When using a variable step size, a minimum integration step size of 1 ms and a maximum step size of 2 s are recommended.

3 Model Configuration

The rated power and dispatched power can be adjusted as explained in the following sub-chapters.

There are different WECC DER models available representing either DER for generation or for storage, as explained in Chapter 1. Both the DER for generation and the DER for storage are represented in the network model via the static generator element. Please note the different settings for generation or for storage on the *Operational Limits* tab of the *Load Flow* page. Further settings with regard to the *DER_A Distributed Energy Resources Model* are described in Section 4.1.

3.1 Adjusting the Rated Power

The rated power of the DER has to be configured on the *Basic Data* page of the DER element itself, i.e. the static generator. The rated active power is calculated via the nominal apparent power “sgn” and the power factor “cosn”. When adjusting the rated power of the DER, the active and reactive power operational limits on the *Operational Limits* tab of the *Load Flow* page may need to be revised.

3.2 Adjusting the Power Output

The power output of the aggregated DER in steady-state load flow calculation can be entered on the *Load Flow* page of the static generator element. The default input mode is “P, cos(phi)”, which allows to enter the dispatched active power and power factor. The reactive power can be limited by constant limits of by a reactive power capability curve which can be also voltage dependent.

3.3 Dynamic Model Configuration

The dynamic simulation is initialised from steady-state, which is the load flow result. It is therefore important to check the load flow configuration of the elements as mentioned in the previous sections.

The dynamic model can either be accessed via the static generator element in the single line diagram or via the data manager.

- Via Element: Open the element. Go to the *Basic Data* page and click on the arrow button after “Model”.
- Via Data Manager: Search the corresponding composite model, right click and select “Edit”.

After opening the composite model, the different dynamic models (DSL models) can be accessed via a double click on the 2nd column. The dynamic models (DSL models) can be configured via scalar input parameters. After opening a DSL model, these parameters are displayed on the *General* page.

The model comes with a reasonable default parameter set. Nevertheless, the parameters of the dynamic model may have to be adjusted by the user in order to reflect behaviour compliant to specific grid connection (grid code) requirements, for example as listed in Table 2.1 of [1].

Some parameters defining the frequency-dependent model behaviour are entered in Hz (parameters *fl* and *fh* in the "DER_A Distributed Energy Resources Model"), and have to be adjusted accordingly with respect to the power system's nominal frequency. Corresponding model templates are provided for use in 50 Hz or 60 Hz power systems.

Please also find the information provided in Chapter 4 with respect to model parameters.

Note: The Static Generator (ElmGenstat) or Photovoltaic System (ElmPvsys), which is used for this model and linked to the DER_A model via the "Frame WECC DER", must have its built-in *Voltage source* model for RMS simulation activated to correspond to the DER_A specification by WECC [3], see Section 4.2.

Note: The parameter values entered for series reactance (*Xseries*) and series resistance (*Rseries*) in the "Voltage Source Reference" model have to correspond to the series reactor settings entered in the static generator (Simulation RMS page of the static generator), see Section 4.2.

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4 Model Documentation

The DER model is represented in the network model via the static generator element. The settings for the steady-state operating point in load flow calculation, which define at the same time the initial conditions for the dynamic simulation, are done in the static generator element. The configuration of the network element is described in Chapter 3. The following sections describe the DSL models which mainly define the model behavior in dynamic simulation in more detail. Table 4.1 lists the DSL models which are included in the DER templates.

Table 4.1: Overview of DSL Models used in the WECC DER Templates

DSL Model	Block Definition (DSL Model Type)
DER_A Distributed Energy Resources Model	DER_A
Voltage Source Reference	Voltage Source Reference (dq)

4.1 DER_A Distributed Energy Resources Model

The WECC DER_A model is an aggregated representation of distributed energy resources for use in dynamic RMS simulation with balanced network representation (positive sequence model). The model is implemented according to the documentation in [3] and [2]. For the graphical representation of the block diagram of the model implementation, refer to the graphic of the DSL Model or DSL Model type within *PowerFactory*. The default parametrisation is based on data given in [2] and [1]. For those parameters values which are not identical in both references, a reasonable selection has been made. The parameters values may have to be adjusted by the user in order to reflect behaviour compliant to specific grid connection (grid code) requirements, for example as listed in Table 2.1 of [1].

In the model, the unit type is configured as generator or storage by using the *typeflag* parameter. In this context, the model parameter *Pmin* can be entered in accordance with the mode of operation.

With regard to the dynamic voltage support functionality, the lower and upper voltage deadband values (*dbd1*, *dbd2*) can be specified related to the voltage deviation of the low-pass-filtered terminal voltage *Vt* (filter time constant *Trv*) from either the initial terminal voltage or a user-defined voltage reference set-point *Vref0*. If *Vref0* is equal to or smaller than zero, the initial terminal voltage is used as reference. The initial terminal voltage is also used as reference, if the deadband is zero (*dbd1* = *dbd2* = 0) or if the deviation of initial terminal voltage from the user-defined *Vref0* input is larger than the configured voltage deadband (initial condition $dbd1 < Vref0 - Vt < dbd2$ is not met). The lower value of the deadband *dbd1* refers to overvoltage events and the upper value *dbd2* refers to undervoltage events, since the voltage deviation is specified as *Vref0 – Vt* (low-pass-filtered) according to [3].

Moreover, the model contains a frequency-power control which can be activated by the parameter *Freq_flag = 1*. In this case, the active power reference is adjusted linearly with the frequency deviation beyond a frequency deadband, which is defined by the parameter *fdbd1* and *fdbd2*. In accordance with [3], the frequency deviation is defined as the deviation of the low-pass-filtered frequency *f* (filter time constant *Trf*) from the reference frequency *fref* (nominal frequency), in the form *fref – f*. If the resulting frequency deviation is greater than the upper frequency control deadband *fdbd2*, i.e. in cases of *under-frequency*, the active power reference is increased linearly with the frequency deviation from upper deadband *fdbd2* multiplied by the the droop gain *Dup*. In contrast, in cases of *over-frequency*, the active power reference is linearly reduced with frequency deviation from the lower deadband *fdbd1* multiplied by the droop gain *Ddn*, if the

calculated frequency deviation exceeds the lower frequency control deadband *fdbd1*.

The parameters for the frequency break point for lower frequency cut-out *fl* as well as for higher frequency cut-out *fh* can be provided in Hz or in p.u. Accordingly, these parameters can take into account the nominal frequency of the power system. Values larger than 10 are interpreted by the model as values in Hz. Values smaller or equal to 10 are interpreted as values in p.u. (based on the network's nominal frequency).

4.2 Voltage Source Reference

The *Voltage Source Reference* model is used to convert current reference signals for a controlled current source model into reference signals for a controlled voltage source model. As the static generator (ElmGenstat) supports both models (controlled current source and controlled voltage source model) this DSL model allows to switch the static generator from one to the other internal model. In the case of the WECC DER_A model, the corresponding static generator needs to be configured using the *Voltage source* model that can be selected on its *Simulation RMS* page.

For the graphical representation of the block diagram of the model implementation, refer to the graphic of the DSL Model or DSL Model Type within *PowerFactory*.

The values of the input paramters *Xseries* (Series reactance in %) and *Rseries* (Series resistance in %) need to match the series reactor settings accordingly that can be entered on the *Simulation RMS* page of the static generator.

The voltage angle detection has a PT1-behaviour (first order lag) with time constant *Tpll*, which is necessary to support *PowerFactory*'s A-stable integration algorithm. The voltage angle detection signals are frozen (keep their previous values) if the terminal voltage is below the value of the parameter *ufreeze*.

Note: The value of the time constant *Tpll* should be selected to represent typical PLL settings. Small values may require sufficiently small simulation time steps in cases of dynamic events, the use of the *Automatic step size adaptation* for the simulation is recommended. Too large values of *Tpll* may result in a too optimistic model behaviour with respect to power system stability.

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5 References

[1] Reliability Guideline, Parameterization of the DER_A Model. Technical report, North American Electric Reliability Corporation, September 2019.

[2] The New Aggregated Distributed Renewable Energy Resources (der_a) Model for Transmission Planning Studies: 2019 Update. Technical Report 3002015320, EPRI, March 2019.

[3] P. Pourbeik. Proposal for a DER_A model 10/11/16 (revised 11/16/16; 3/6/17, 3/15/17; 3/28/17; 3/29/17; 3/31/17; 4/17/17; 10/5/17; 11/9/17; 2/9/18; 2/15/18; 3/9/18; 7/17/18; 8/29/18; 9/11/18). 2018.

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